



TheLook E-commerce Growth & Retention Rate

Analysis Using SQL and Looker
Studio

By Ashish Kumar

Table of contents

01. About company

02. Problem Statement

03. Methodology

04. Dataset Overview

05. Data Modelling

06. Insights & Recommendation

07. Dashboard

Creator



Ashish Kumar

Data Analyst



About company

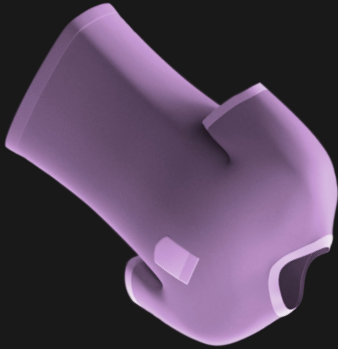
The Look is a fictitious e-Commerce clothing site developed by the Looker team. The dataset contains information about customers, products, orders, logistics, web, events, and digital marketing campaigns.

The contents of this dataset are synthetic and are provided to industry practitioners for the purpose of product discovery, testing, and evaluation.



Problem Statement

I'm currently employed as a data analyst in the data analytics department of The Look, an online retailer of clothing, and part of my job description is to share insights regarding the company's products, operations, and marketing with the appropriate parties so that the enterprise may be sustained (expanding and profitable). Due to a probable crisis in 2023, the company is in optimization mode. The management made the decision to cut off resources in some of the categories with the worst growth over the previous year. In order to increase the retention rate, they also aim to continue their analysis by learning about user retention behaviors.



Methodology

We will use **Look ML** to model data, **SQL (Google Big Query)** to answer questions, and **Looker Studio** to visualize the results of this investigation.



Google
Big Query



Dataset Overview

Dataset info

 [EDIT DETAILS](#)

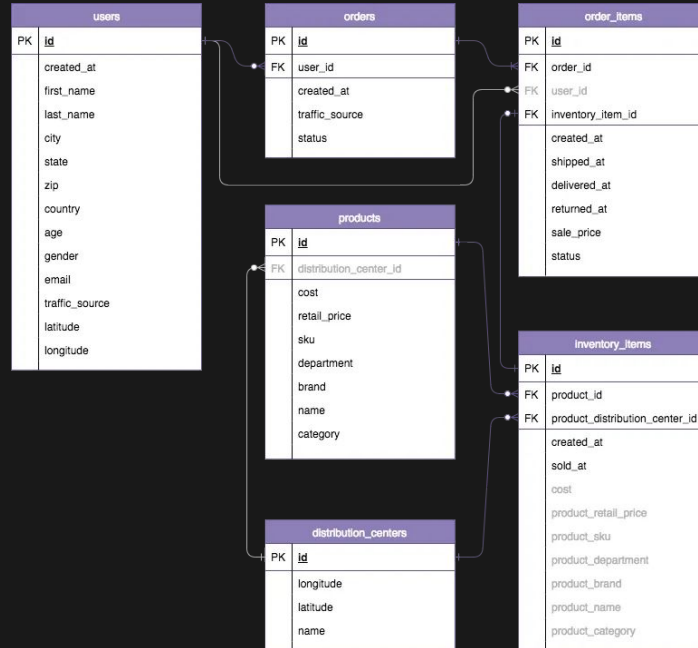
Dataset ID	bigquery-public-data.thelook_ecommerce
Created	Feb 26, 2022, 5:22:02 AM UTC+5:30
Default table expiration	Never
Last modified	Sep 20, 2022, 1:28:24 PM UTC+5:30
Data location	US
Description	Fictitious E-Commerce Dataset
Default collation	
Case insensitive	false
Labels	

▼	 thelook_ecommerce	☆	⋮
	 distribution_centers	☆	⋮
	 events	☆	⋮
	 inventory_items	☆	⋮
	 order_items	☆	⋮
	 orders	☆	⋮
	 products	☆	⋮
	 users	☆	⋮

Schema Description

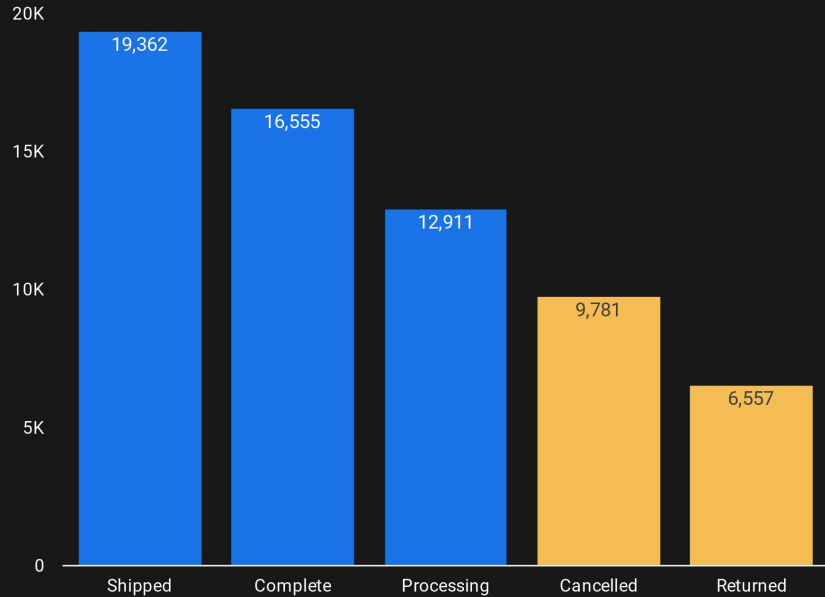
distribution centers	store information about various distribution centers, including their identifiers, names, and geographic coordinates.
events	store various types of event data related to user interactions or activities.
Inventory items	manage and track inventory levels, sales, and associated product information.
order items	manage and track individual items within orders, including user information, product details, fulfillment status, timestamps, and pricing.
orders	manage and track individual orders, including user information, order status, timestamps related to different stages of the order process, and the quantity of items in each order.
products	manage and track information about individual products, including their characteristics, pricing, and distribution center affiliation.
users	manage and store information about individual users, including their personal details and contact information.

Data Modeling



Growth Analysis

Orders Status



Insights

- From 65,166 orders, 15% were cancelled and 10% were returned.

Recommendation

- We must check the reason behind the cancelled orders; are there any difficulties with our order or payment funnel?
- We should also address the issue of returned orders: is there an issue with our courier? Or is there a problem with our products?

Customers with the Highest Order Returns

	id	email	first name	last name	count ▾
1.	1881	ryanespinoza@example.net	Ryan	Espinoza	2
2.	15350	markgonzales@example.com	Mark	Gonzales	2
3.	57319	karenunderwood@example.net	Karen	Underwood	2
4.	79910	tristanmorrison@example.net	Tristan	Morrison	2
5.	30211	anamack@example.org	Ana	Mack	2
6.	25624	vincentlee@example.org	Vincent	Lee	2
7.	89904	lynndeleon@example.org	Lynn	Deleon	2
8.	30123	stevenparker@example.net	Steven	Parker	2
9.	62894	gabriellaellis@example.com	Gabriella	Ellis	2
10.	9551	marisaramirez@example.net	Marisa	Ramirez	2

Grand total 2,332

1 - 10 / 2299 < >

Insights

- Total of 2332 orders ordered return from 2299 users.
- These are the 10 people who returned their orders the most.

Recommendation

- We can investigate or follow up with these people to find out why their orders were returned.

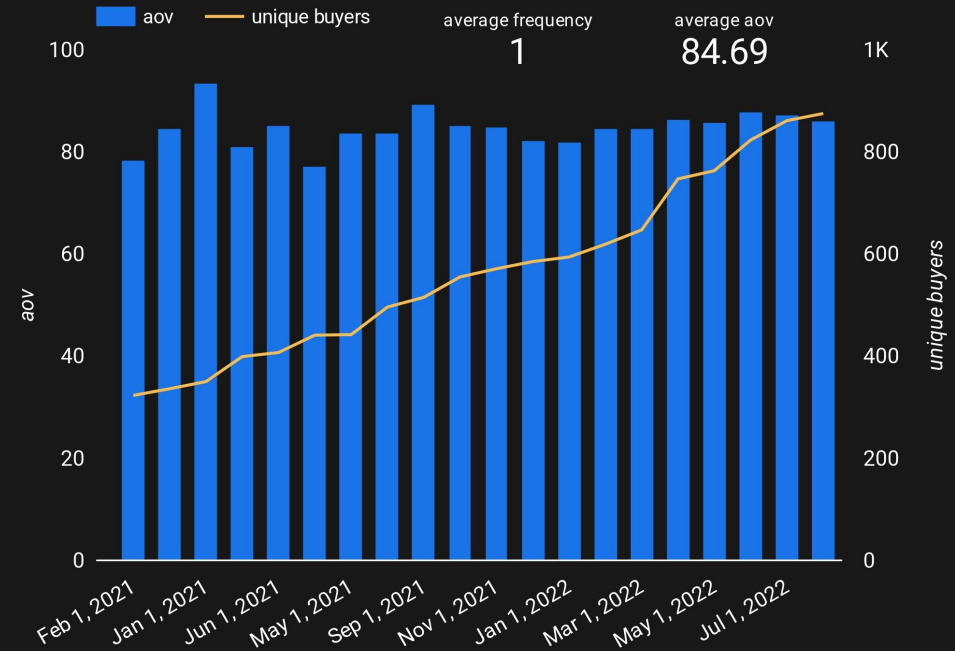
Customers Purchasing Patterns

Insights

- The average number of orders for every user in a month is **1** and average sale price per order is **\$84.69**.
- The number of buyers is increasing over the years, but we can see that the total sale price per order stays the same, which means buyers tend to buy at a lower price.

Recommendation

- Implement customer loyalty programs that reward frequent buyers with discounts, exclusive offers, or early access to new collections.
- Display related or complementary items when customers are viewing a product



Most Profitable Items

	id	name	cost	retail price	profit ▼
1.	23648	The North Face Apex Bionic Jacket - Men's	\$399.13	\$903.00	\$4,534.87
2.	24341	Nobis Yatesy Parka	\$381.90	\$950.00	\$3,976.70
3.	2557	Canada Goose Women's Mystique	\$280.50	\$750.00	\$3,756.00
4.	8774	Nobis Tula Parka	\$364.65	\$825.00	\$3,222.45
5.	20227	Joseph Abboud Men's 2 Button Side Vent Suit Wi...	\$158.46	\$417.00	\$3,102.48

Least or Most: most profitable (1) ▼

Least Profitable Items

	id	name	cost	retail price	profit ▲
1.	14235	Indestructable Aluminum Aluma Wallet - RED	\$0.01	\$0.02	\$0.06
2.	12664	Skyblue Plain Color Cotton Ribbon Adjustable El...	\$1.54	\$2.99	\$1.45
3.	12536	Individual Bra Extenders	\$1.01	\$1.75	\$1.47
4.	28548	HDE Classic Bow Tie	\$1.28	\$2.95	\$1.67
5.	14159	Set of 2 - Replacement Insert For Checkbook Wall...	\$0.18	\$0.49	\$1.88

Least or Most: least profitable (1) ▼



Nobis Yatesy Parka

Most Profitable

- The total opposite of the most profitable products, these least profitable products have low costs, so their retail prices are also low.
- The profit came from selling 1-2 items.
- These products must be removed because they didn't sell well.

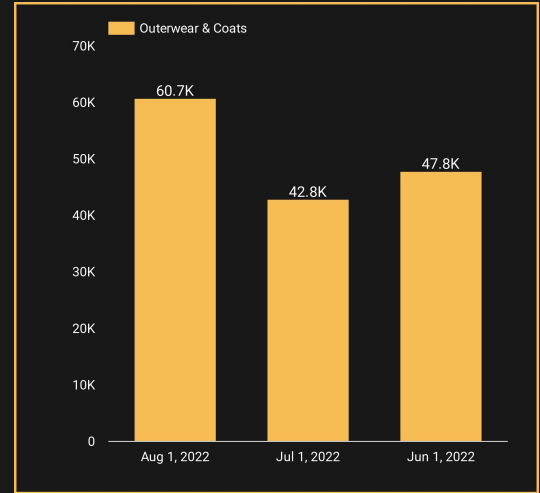
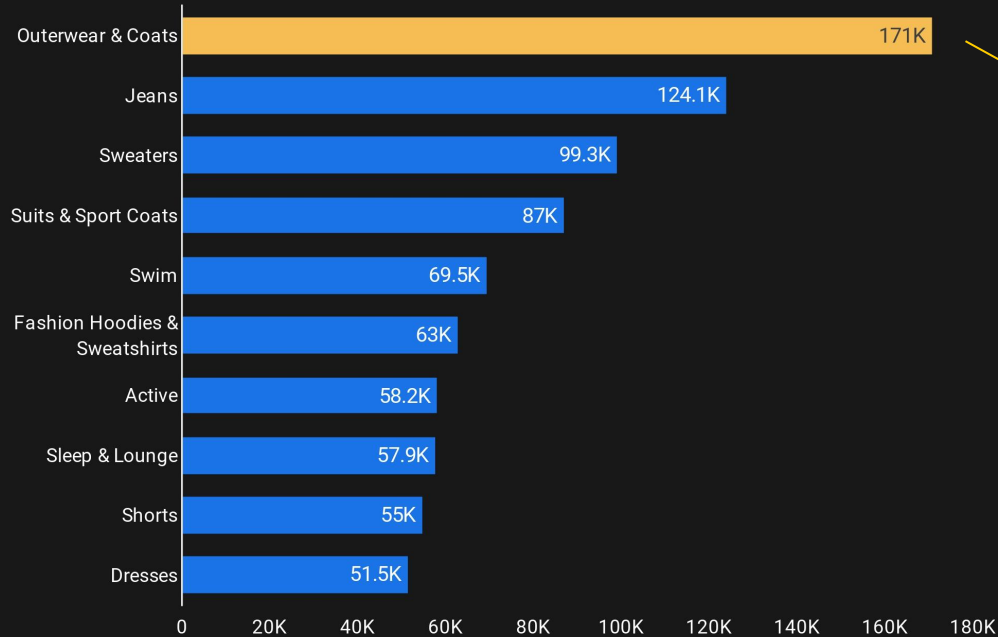
- From the most 5 profitable items, we could see that most of them have high price, hence the high profit.
- These products get their profit selling around 10 items.
- We could stock these products more or spend our marketing budget to increase sales.



Aluminum Aluma Wallet

Least profitable

Profit by Category



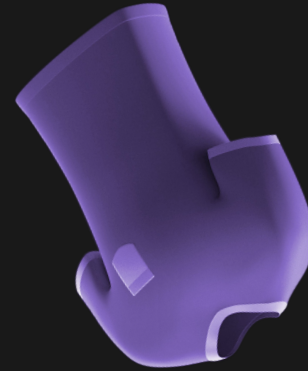
* Profit figure in dollar(\$).

Insights

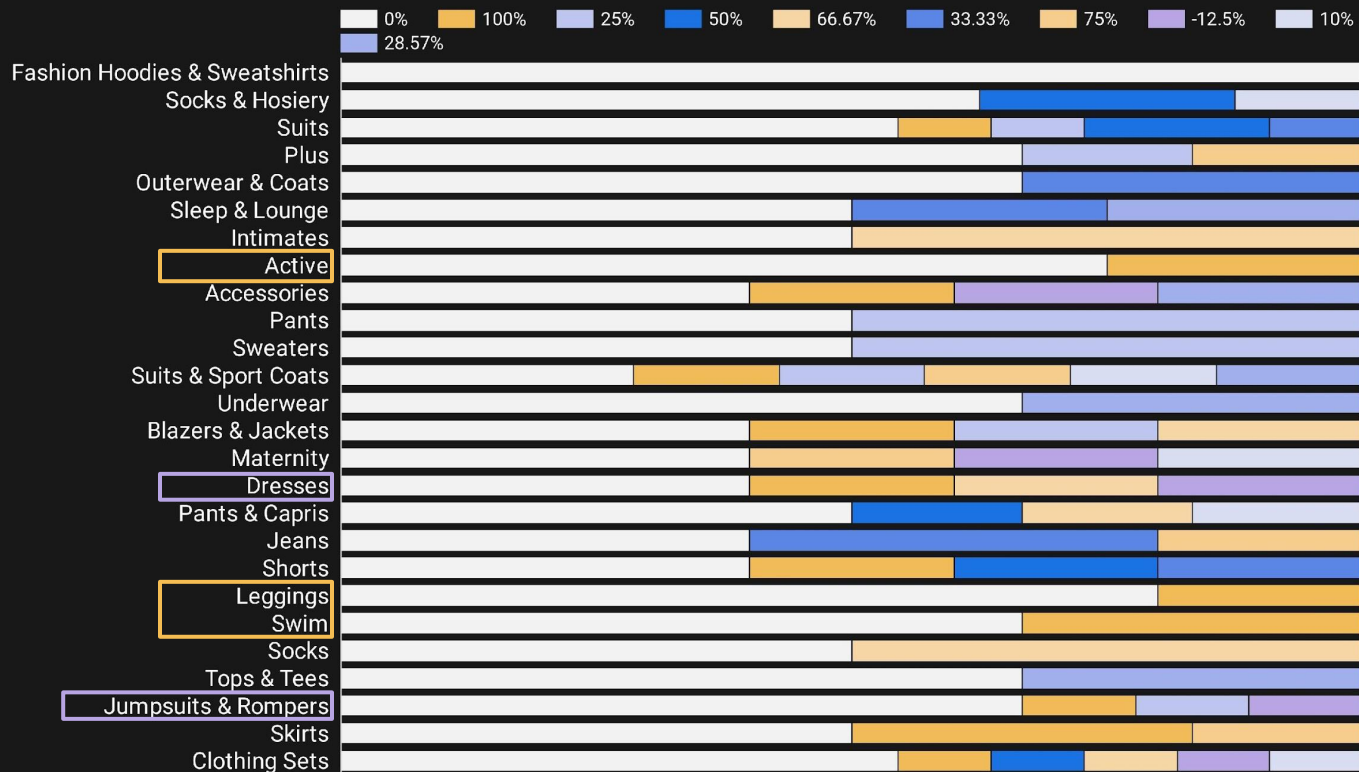
- Outerwear and Coats has been the most profitable category in the past 3 months.
- Its highest profit during the past 3 months was in August, which was **\$60.7k**.

Recommendation

- Increase Outerwear and Coats inventory and add more diverse products



Product Growth

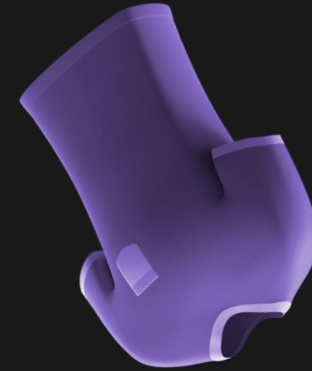


Insights

- Swim, Active, and Leggings have the fastest growth of all categories.
- Jumpsuits and Rompers have the lowest growth of the all categories.

Recommendation

- Focus marketing efforts on Swim, Active, and Leggings categories, as they are experiencing the fastest growth.
- Since Jumpsuits and Rompers exhibit the lowest growth, consider diversifying your offerings within these categories. Explore new designs, styles, or collaborations that could attract a wider audience and renew interest in these products



Cohort Analysis

Customers Retention

month number / percentage								
cohort month	cohort size	0	1	2	3	4	5	6
Jan 1, 2022	1686	100%	3.02%	3.2%	2.79%	2.91%	3.2%	3.38%
Feb 1, 2022	1626	100%	3.26%	2.95%	4.12%	3.44%	3.32%	3.63%
Mar 1, 2022	1794	100%	3.29%	4.01%	3.4%	3.9%	3.62%	3.46%
Apr 1, 2022	1783	100%	3.53%	3.14%	4.09%	3.87%	3.76%	4.15%
May 1, 2022	1937	100%	3.82%	4.03%	4.28%	4.49%	4.44%	3.56%
Jun 1, 2022	1989	100%	3.37%	4.58%	4.32%	4.12%	3.92%	4.02%
Jul 1, 2022	2107	100%	4.7%	4.37%	4.75%	4.32%	3.7%	5.6%
Aug 1, 2022	2213	100%	4.16%	4.02%	4.02%	4.25%	3.93%	4.2%
Sep 1, 2022	2284	100%	4.9%	4.9%	5.04%	4.82%	5.08%	4.51%
Oct 1, 2022	2416	100%	4.51%	5.34%	4.43%	4.59%	4.72%	4.72%
Nov 1, 2022	2450	100%	6%	5.88%	4.53%	4.57%	5.35%	5.43%
Dec 1, 2022	2647	100%	6.95%	5.4%	5.97%	5.63%	6.16%	5.18%
Grand total		1,200%	51.52%	51.82%	51.74%	50.91%	51.2%	51.84%

Customers Retention

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May 1, 2022	1937	100%	3.82%	4.03%	4.28%	4.49%	4.44%	3.56%
Jun 1, 2022	1989	100%	3.37%	4.58%	4.32%	4.12%	3.92%	4.02%
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Grand total		1,200%	51.52%	51.82%	51.74%	50.91%	51.2%	51.84%

1.

Increasing cohort size (users) with each new month.

Customers Retention

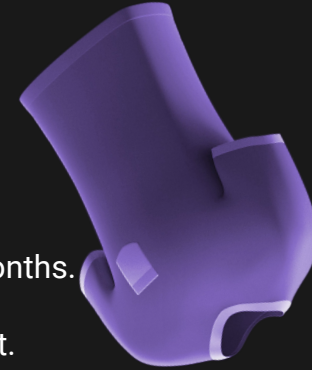
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Aug 1, 2022	2213	100%	4.16%	4.02%	4.02%	4.25%	3.93%	4.2%
Sep 1, 2022	2284	100%	4.9%	4.9%	5.04%	4.82%	5.08%	4.51%
Oct 1, 2022	2416	100%	4.51%	5.34%	4.43%	4.59%	4.72%	4.72%
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Dec 1, 2022	2647	100%	6.95%	5.4%	5.97%	5.63%	6.16%	5.18%
Grand total		1,200%	51.52%	51.82%	51.74%	50.91%	51.2%	51.84%

2.

Noticeably decreasing retention percentage.

Insights

- Positive trend in user growth as each cohort start with more users.
- Retention percentage show consistent stability across different cohort months.
- Challenging with retaining users after 4 - 6 months are evident in all cohort.
- December 2022 cohort's retention decrease could be an anomaly due to data endpoint.

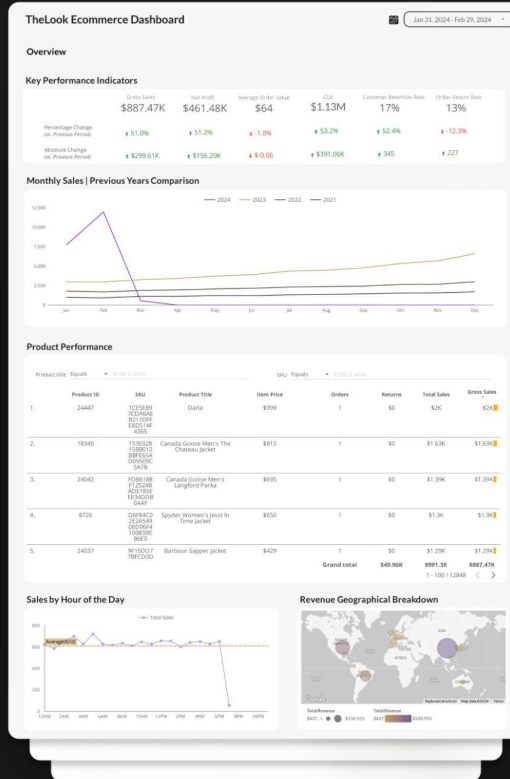
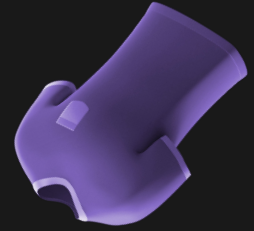


Recommendation

- Boost engagement during 4 - 6 months by offer personalized recommendations and exclusive offers.
- Gather user feedback and conduct research to uncover reasons for attrition after 4 - 6 months.
- Utilize A/B Testing to refine engagement strategies.
- Extend data collection to understand true retention trend for December 2022 cohort.



Dashboard



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